

The empirical recurrence for OEIS A223594, proved by a two-line transfer matrix

Adrian Perez Fontelles
Independent researcher

31 August 2026

Abstract

OEIS A223594 counts arrays over $\{0, \dots, 15\}$ in which every cell is constrained by the values of its neighbours, and carries an empirical recurrence of order 7 contributed by R. H. Hardin. It is true, and it is not an empirical matter. The neighbourhood reaches one row up and one row down, so a single row is not enough state and a plain walk count is wrong; the right object is a walk on ordered *pairs* of consecutive rows, where each step tests the condition on the middle row against both its neighbours and the two ends are tested with one neighbour row absent. That makes $a(n)$ a walk count on a digraph with $S = 4096$ vertices, and the conjectured recurrence is then decided by a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The sequence and the conjecture

OEIS A223594 is “Petersen graph (8,2) coloring a rectangular array: number of $n \times 3$ $0..15$ arrays where $0..15$ label nodes of a graph with edges $0,1$ $0,8$ $8,14$ $8,10$ $1,2$ $1,9$ $9,15$ $9,11$ $2,3$ $2,10$ $10,12$ $3,4$ $3,11$ $11,13$ $4,5$ $4,12$ $12,14$ $5,6$ $5,13$ $13,15$ $6,7$ $6,14$ $7,0$ $7,15$ and every array movement to a horizontal, diagonal or antidiagonal neighbor moves along an edge of this graph.”. It has offset 1 and begins

144, 1504, 16192, 176224, 1931968, 21308000, 236213312, 2629972704, ...

The entry carries the following comment:

Empirical: $a(n) = 23*a(n-1) - 153*a(n-2) + 217*a(n-3) + 258*a(n-4) - 456*a(n-5) - 104*a(n-6) + 192*a(n-7)$.

As of the “Last modified” line on the live entry (Aug 21 2018 10:04:26, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 23a(n-1) - 153a(n-2) + 217a(n-3) + 258a(n-4) - 456a(n-5) - 104a(n-6) + 192a(n-7) \quad (1)$$

for all $n > 7$.

2 The condition, and why one row is not enough

The arrays have 3 columns and $L = n$ rows; write $x_{t,u}$ for the entry in row t and column u , each in $\{0, \dots, 15\}$. With the neighbour offsets

$$\mathcal{N} = \{(0, 1), (1, 1), (1, -1)\},$$

Proof. Substitute Lemma 1 and factor G^{j-7} out of $G^j - \sum_i c_i G^{j-i}$; the nonzero scalar 1 does not affect vanishing. For the last clause, Cayley–Hamilton makes G^S an integer combination of $I, G, \dots, G^{S-1}$, so $\mathbf{v}^\top G^m w$ for $m \geq S$ is the corresponding combination of the earlier values; if those vanish, so do all the rest. $\square$

Theorem 3. *The recurrence (1) holds for every $n > 7$, which is the statement of Section 1.*

Proof. The digraph was built directly from (2): for each of the $S = 4096$ pairs and each of the $(15+1)^3$ possible next rows, the condition was tested on every cell of the middle row. The vector w was formed by 7 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots$, again exactly. Every one of those integers vanishes from the index corresponding to $n = 7 + 1$ onwards, and the run was continued until $S + 1$ consecutive zeros had been seen, which by Lemma 2 settles every larger m at once. $\square$

Remark 4. The argument gives more than the recurrence: the sequence is C -finite with characteristic polynomial dividing that of G , hence has a rational generating function whose denominator divides $\det(I - xG)$. The conjectured recurrence is one consequence; the minimal one is a divisor of q .

5 Verification

Three checks, each independent of the algebra above.

First, the walk counts of Lemma 1 were compared against the entry’s DATA at every one of the 16 published indices, with no shift fitted: the number of rows is read off the entry’s own name as $L = n$ and the entry’s offset says which index the first published term carries. The values agree exactly. A misreading of (2), of the neighbour set, or of the boundary treatment would give a different digraph and different counts, so this is a test of the modelling and not only of the arithmetic.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 2 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix, and returned zero throughout.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A223594.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.